Application of a Fuzzy ARTMAP Neural Network for Indoor Air Quality Prediction

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***Abstract*— Indoor air quality monitoring is an important activity to ensure continued health and well-being of citizens living, studying, and working in indoor environments. This practice has been widely developed through the application of low-cost sensors that are able to measure gas concentrations, particulate matter, and other components such as humidity and temperature that affect indoor air quality. Additionally, machine learning algorithms have been applied in the interpretation of sampled environmental data to improve the performance of monitoring systems. This paper proposes the implementation of a fuzzy ARTMAP neural network, which employs the concepts of Adaptive Resonance Theory (ART), to compute the prediction of particulate matter sampled in a domestic bedroom environment. With the application of a specialized online training architecture, the fuzzy ARTMAP network can be a promising alternative to predict particulate matter time series data modeled in sliding windows, obtaining predictions 24-hour ahead with mean absolute error (MAE) ranging here from 0.26 to 7.65.**

Keywords—indoor air quality, particulate matter prediction, fuzzy ARTMAP neural network, online training

I. Introduction

Technology is increasingly present in everyday life and has become fundamental to promote improvements in the quality of life within modern society. However, constant technological revolution has triggered the uncontrolled consumption of natural resources, and one of the consequences of this process is increasing environmental pollution. Among the types of pollution, poor air quality is a major threat to health and ecosystems [1]. Air quality study has become one of the most discussed environmental concerns [2].

Some of the typically considered air monitored pollutants to affect human health are ozone (O_3), particulate matter (PM), carbon monoxide (CO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), lead (Pb), radon gas (Rn), and organic volatile compounds (VOC). To classify air quality as safe, in addition to ranges stipulated for pollutant concentrations, it is necessary to determine a maximum exposure period for an analyzed scenario [3].

Factors that have contributed to the study of air quality are the development of low-cost sensors and an increase in access to tools needed to analyze the sampled compounds [4]. To improve the interpretation of sampled data, an alternative to traditional approaches that is increasingly used is machine learning (ML). With the knowledge acquired by the interaction of the computational algorithm with environmental stimuli, mathematical models can recognize existing patterns and behaviors, enabling the realization of intelligent inference and prediction in observed contexts.

Regarding the monitored air quality environment, it is common to be only concerned with outdoor air quality [5]. However, controlling indoor air quality (IAQ) is equally important. The immediate need to study IAQ is due to the tendency of people to stay indoors for up to 90% of the time [6]. Examples of indoor environments include domestic (home), workplace (office), public buildings (such as hospitals, schools, theaters, libraries, and restaurants), and vehicles [7].

One of the main air pollutants monitored in both outdoor and indoor environments is particulate matter (PM) [8]. Indoors, PM can cause irritation to the eyes, nose, and throat, worsening symptoms of respiratory diseases, and premature death in people with heart or lung diseases [9].

The importance of analyzing PM concentrations in internal environments can be seen by the number of inhabited spaces. An association of ML algorithms and indoor air quality monitoring systems was proposed in [10] with the application of a Discrete-Time Markov Chain to predict IAQ in a university environment involving air compounds that included PM. Another interesting possibility for monitoring indoor university environments was carried out in [11], who evaluated a number of ML algorithms, such as Support Vector Machine (SVM) and Radial Basis Function (RBF) Kernel, to predict maximum occupancy of these environments aiming at maintaining air quality based on current IAQ indexes.

PM was also monitored inside a restaurant in [12]. It is also possible to study how the concentration of PM in internal environments is influenced by several other factors. For this, [13] proposed a multiple linear regression model to identify that external pollution levels and human actions, such as opening and closing windows in certain periods, modify the indoor PM concentration. Thus, monitoring indoor pollutants and applying ML algorithms for processing sampled data in real-time are promising tools to obtain systems capable of keeping users informed about the state of the air quality, allowing them to perform interventions to keep the compounds within safe levels stipulated by the regulatory entities. Given the presented scenario, this paper proposes to implement and evaluate the performance of an artificial neural network (ANN) which applies the concepts of Adaptive Resonance Theory (ART), more specifically the fuzzy ARTMAP neural network, to calculate the prediction of indoor air quality time series data. As a case study, the

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indoor pollutant considered in this paper was particulate matter.

This paper is organized as follows. In Section II, the basic ART concepts are presented, including a modified fuzzy ARTMAP neural network architecture to improve online training. In Section III, the proposed system implementation is detailed. Section IV presents and discusses the results, and Section V provides conclusions.

II. Improved Fuzzy ARTMAP Neural Network to Perform Online Training

Adaptive Resonance Theory (ART) was proposed by Stephen Grossberg in 1976 and is a mathematical model built on two dilemmas: plasticity and stability. As the model interacts with external agents, plasticity indicates the ability to learn new patterns and place them in specific clusters. Stability on the other hand guarantees the preservation of earlier learned information whilst distinguishing new patterns produced from current data [14].

An important architecture within the ART neural network family was generated with the association of fuzzy logic operators [15] to obtain the fuzzy ART architecture. This model can process analog and binary data [16] by applying unsupervised training to group the input patterns into clusters.

Additionally, the fuzzy ARTMAP network [17] was proposed by combining two fuzzy ART modules, named *ARTa* and *ARTb*, which are responsible respectively for clustering the input and output data. The use of a mapping method implemented within the *Inter-ART* module combines the activity of the fuzzy *ARTa* and fuzzy *ARTb* modules, and the fuzzy ARTMAP neural network has stable and incremental learning through supervised training.

Although traditional fuzzy ARTMAP supports online training, its standard architecture can be modified to optimize information processing during continuous learning. One possible improvement is to manage the cluster creation, as proposed by [18], separating the A*RTa* clusters into two groups: temporary clusters and definitive clusters. The newly formed *ARTa* clusters are considered temporary information. The criterion for a temporary cluster to become definitive is that it be selected a minimum number (*Nmin* parameter) to represent input patterns during the training process. On the other hand, temporary clusters that have not been selected to classify input patterns after the network receives a certain number of input patterns (*Nmax* parameter) are excluded, as they are considered irrelevant information.

With this new cluster management methodology, the fuzzy ARTMAP knowledge is constantly updated, maintaining the most relevant information in the definitive cluster, and the memory usage is improved due to the temporary cluster exclusion.

The fuzzy ARTMAP architecture diagram considering the modifications proposed in [18] is shown in Fig. 1. The *ARTb* module contains the standard structure of fuzzy ART, which is comprised of the F_0^b layer, where pre-processing operations are performed, F_1^b layer, which is an interface that connects the input to the output layer, F_2^b. The artificial neurons of this ANN are stored in the w_k^b weight matrix. In the *ARTa* module, there is cluster separation in the F_2^a layer between the temporary (w_j^t) and definitive (w_j^d) weights. Mappings between temporary and definitive input clusters with output clusters are stored respectively in the w_j^{abt} and w_i^{abd} matrices, which are updated by the Match tracking operations processed in the *Inter-ART* module.

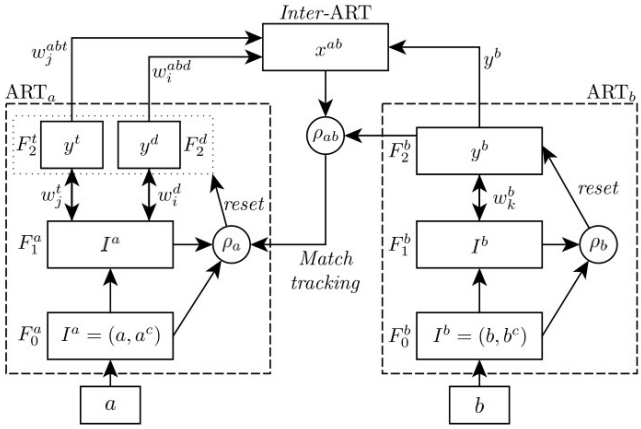

Fig. 1. Fuzzy ARTMAP neural network architecture.

With online training and the application of temporary and definitive categories, the fuzzy ARTMAP neural network could be a good candidate to perform real-time prediction. The fuzzy ARTMAP neural network implementation carried out to predict PM time series data is shown below.

III. Sensor System Development

A low-cost PMS7003 PM sensor was used to perform IAQ monitoring in a domestic bedroom environment where one person resides. The PMS7003 air quality sensor, as shown in Fig. 2, is a laser scattering sensor that can detect air suspending particles with different diameters [19]. In this paper, particulate matter with diameter of 2.5 micrometers (PM2.5) was considered.

Fig. 2. PMS7003 particulate matter sensor.

The sensor was placed about 50 centimeters from the floor. The data sampling was carried out in continuous PMS7003 operation mode, which performed readings at a rate of approximately 1 Hz. A Raspberry Pi 3 managed the communication with the sensor and stored the samples in a cloud database service. The data processing was made after the sample phase by a Python script executed on a laptop computer. The goal of this script was to organize the discretized PM2.5 sampled data to be presented to the fuzzy

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ARTMAP neural network. The time series was obtained with the 1-hour data mean, resulting in 24 values per day.

To apply the fuzzy ARTMAP network and predict PM2.5 time series data, it was necessary to organize the continuous sensor sampled data for a supervised training approach.

In this paper, fixed size overlapping sliding windows were used. To generate a window at each instant t of the discretized time series, three previous lag values were considered. The number of lags defines the overlap size of adjacent windows. In addition to the discretized PM2.5 values represented by the variable v, it was necessary to add historical information to the input patterns so that the fuzzy ARTMAP network performs the classification based on the variable values and corresponding time information. The considered time data were the day of the week and the hour of day, respectively, represented by the variables d and h that are present in the vector t described by (1), which is the fuzzy ARTMAP input:

$$a = [d, h, v(t\text{-}n), \ldots, v(t\text{-}1), v(t)] \quad (1)$$

The fuzzy ARTMAP output is the v variable value at the subsequent input instant, as represented by (2):

$$b = [v(t+1)] \quad (2)$$

The proposed data modelling approach allows the prediction calculation one instant ahead of each sample, as defined in (2). However, it was desired to obtain predictions for a longer future time period. For this, subsequent predictions were made using the ANN output as its input to build the next input patterns to calculate PM predictions 24 hours ahead.

Each variable was normalized by applying a division by the highest correspondent value to guarantee all values to be transformed in the range 0 to 1, as required by the fuzzy ARTMAP neural network. To obtain the correct prediction value, it is also necessary to undo the normalization, therefore the highest variable values were stored.

Sampling occurred from October 20th to December 11th 2020. It was decided that during the first week of operation, the system was only trained using a period comprised of 168 input patterns. Upon receipt of subsequent six weeks of data totaling 1008 input patterns, training was maintained and prediction calculations began, so characterizing an online training approach. During the fuzzy ARTMAP learning phase, it is necessary to present the input (a) and output (b) vectors to perform the mapping between the input pattern and the corresponding output values. To make predictions, it was only necessary to present the a input vector. The prediction response was obtained considering the *ARTb* output mapped to the active *ARTa* cluster.

IV. Results and Discussion

The fuzzy ARTMAP neural network implementation was performed in software using the Python programming language. The evaluations were run on a laptop computer with an Intel Core i5-8250U processor and 8 GB of RAM. It should be noted that the evaluations were not performed in real-time. Online training was however performed, faithfully reproducing the events that would occur in an application with streaming data sampling.

Table 1 shows the defined fuzzy ARTMAP parameters. Regarding the online training parameters, it was defined that a temporary cluster must be activated twice to become definitive ($Nmax = 2$), and to delete those that remain inactive for two weeks. That is, those that do not classify any input after the network receives 336 inputs ($Nmin = 336$).

TABLE I. Fuzzy ARTMAP parameters

Parameter	Value
Choice parameter (α)	0.7
Training rate (β)	1.0
ARTa baseline vigilance parameter (ρ_a)	0.99
ARTb vigilance parameter (ρ_b)	0.96
Inter ART vigilance parameter (ρ_{ab})	1.0
ρ_a increment	0.0001
Nmax	2
Nmin	336

Fig. 3 shows a plot of online training and performance with the realized real-time fuzzy ARTMAP neural network. In the considered methodology, it was desired to obtain the PM2.5 estimation always 24 hours ahead. Thus, in each sampling phase, the fuzzy ARTMAP training was carried out, and 24 predictions calculated to obtain the constantly updated ahead prediction. Therefore, the organization of successive predictions, as shown in Fig. 3, is equivalent to a sequence of one-hour ahead predictions.

As presented, the fuzzy ARTMAP neural network response was able to extract temporal information from the input data, as the prediction was close to the real data. In

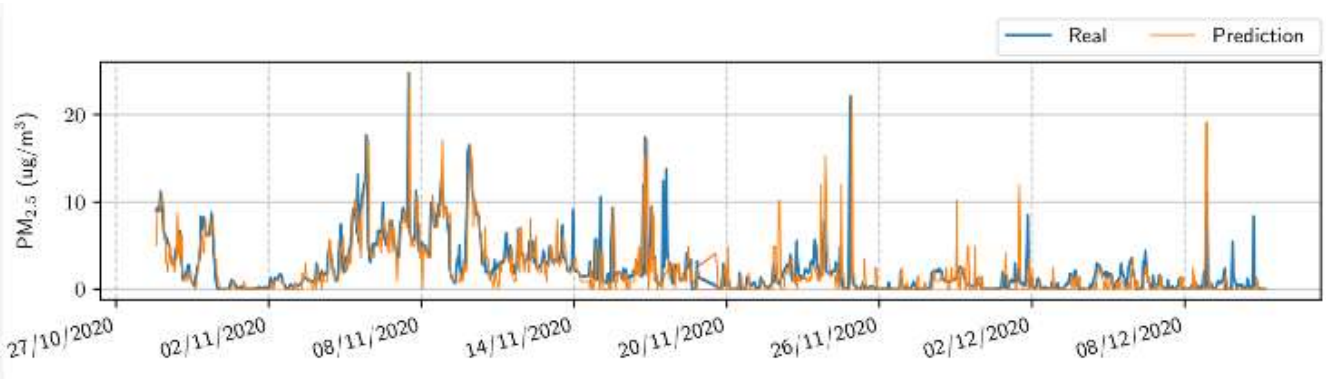

Fig. 3. Fuzzy ARTMAP neural network PM2.5 prediction.

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addition to graphical evaluation, it is possible to measure the prediction accuracy using metric errors. For this, it was considered the mean absolute error (MAE) calculated by equation (3):

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t| \quad (3)$$

where n is the number of calculated predictions, y_t the actual value of the time series at the instant t and $\hat{y}_t$ the fuzzy ARTMAP prediction value. In this case, the MAE was 0.99.

To demonstrate the system operation in real time, that is a complete fuzzy ARTMAP prediction set calculated after a data sample presented as its input, Fig. 4 shows a 24-hour ahead prediction set calculated by the fuzzy ARTMAP neural network.

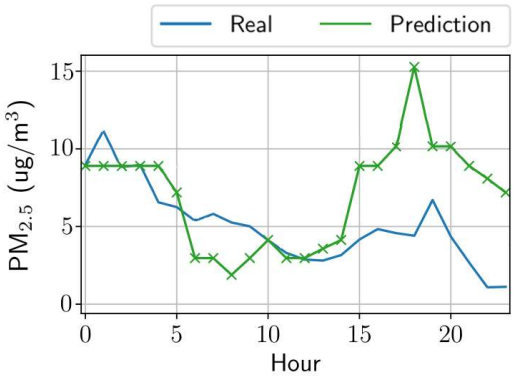

Fig. 4. 24-hour ahead prediction set example plot.

It is also possible to evaluate the prediction accuracy in more detail through the MAEs of each 24 hour ahead prediction set. Table 2 shows the MAEs of highest, lowest, and mean of the 1008 prediction sets of PM2.5.

TABLE 2. MAEs of 24-hour ahead prediction sets calculated by fuzzy ARTMAP.

Highest	Lowest	Mean
7.65	0.26	2.28

With MAE equal to 2.28, the mean accuracy for a 24-hour ahead prediction is higher than the MAE of the general prediction shown in Fig. 3, which was equal to 0.99. As described, to obtain predictions several instants ahead, the fuzzy ARTMAP response is considered as its input to create future input patterns. In this process, network-calculated errors are propagated to subsequent predictions, increasing the loss of accuracy as predictions progress. The accumulated error can be minimized by making fewer subsequent predictions but would lead to a decrease in the temporal prediction range.

V. Conclusions

This paper has proposed the implementation of a fuzzy ARTMAP neural network to perform air quality prediction using PM2.5 time series sampled in a domestic bedroom. With online training and the ability to compute predictions concurrently with data sampling, the system proved to be a good alternative to perform indoor environments air compounds prediction in real-time. It is possible to obtain multiple future predictions with the fuzzy ARTMAP neural network. However, the further the predictions are in the future, the lower the accuracy of the responses obtained. Therefore, it is still necessary to carry out studies in this scope to improve the fuzzy ARTMAP neural network prediction.

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